# OBJECTIVE OF THE WORKSHOP

- **Core Focus: Explore how advanced metallic materials and their microstructural features determine performance in modern engineering applications**
- **Key Learnings: Gain insights into recent advances in alloy design, processing, microstructural characterizations, and mechanical behaviour.**
- **Industry Impact: Understand the vital importance of structure-property relationships in developing stronger, lighter, more durable materials for energy-efficient and sustainable manufacturing.**
- **Methodology: Discover how advanced characterization tools and computational approaches are used to assess microstructure and predict performance through expert talks, interactive sessions, and discussions.**
- **Ultimate Goal: Inspire collaboration and innovation among students, researchers, and industry professionals in metallurgy and materials engineering.**

**NATIONAL INSTITUTE OF TECHNOLOGY JAMSHEDPUR**

**Department of Metallurgical and Materials Engineering**
**Presents**
**A 5-Day workshop on**

# METAMAP 2027

**8TH - 12th January 2027**

**METALLURGY AND MATERIALS: EMERGING TRENDS IN ADVANCED MATERIALS, MICROSTRUCTURAL ASSESSMENT AND PERFORMANCE**

## CONTACT US

### CONVENORS


**Dr. Anushree Dutta**
Assistant Professor
Dept. of Metallurgical and Materials Engineering
NIT Jamshedpur
Contact : +91 80172 65397
Email : anushreedutta.met@nitjsr.ac.in

**Dr. Amit Patel**
Assistant Professor
Dept. of Metallurgical and Materials Engineering
NIT Jamshedpur
Contact : +91 83050 26990
Email : amitpatel.met@nitjsr.ac.in


### COORDINATORS


**Dr. Prakash Sarkar**
Assistant Professor
Dept. of Metallurgical and Materials Engineering
NIT Jamshedpur
Contact : +91 88284 95001
Email : prakash.met@nitjsr.ac.in

**Dr. Jichil Majhi**
Assistant Professor
Dept. of Metallurgical and Materials Engineering
NIT Jamshedpur
Contact : +91 96580 72304
Email : jichilmajhi.met@nitjsr.ac.in




## About the Institute

Founded on August 15, 1960, as the Regional Institute of Technology (RIT), a joint venture between the Government of India and the Government of Bihar. It is one of the original eight Regional Engineering Colleges (RECs) launched under India's Second Five-Year Plan (1956–1961) and historically the sole REC designated as RIT. It was upgraded and renamed as the National Institute of Technology (NIT) Jamshedpur on December 27, 2002. The Institute comprises 12 academic departments spanning engineering, science, and humanities and operates as a self-contained 340-acre campus housing around 4,000 students, 204 faculty members, and 150 administrative and supporting staff.

## About the Department

Established in 1960 as one of the institute's four foundational branches (originally under RIT), the department evolved its curriculum in 1995 to formally unite traditional metallurgy with materials engineering. Situated in Jamshedpur's mineral-rich industrial hub, it specializes in key areas such as Materials and Foundry Technology, Iron and Steelmaking, Nanotechnology, and Mineral Benefaction. Supported by modern experimental and computational facilities, its research spans physical metallurgy, hydrometallurgy, process metallurgy, and mineral characterization. Leveraging deep ties with major industry leaders like Tata Steel, JSW, etc. The department consistently delivers strong placement outcomes and top national academic recognitions.

DATE : 8TH - 12TH JANUARY 2027
VENUE : DJLHC, NIT JAMSHEDPUR

# KEY FEATURES

- Expert Talks
- Cutting-Edge Research
- Industry Insights
- Live-demonstrations
- Networking Opportunities
- Real-World Applications

# WHO SHOULD ATTEND

- Industry professionals
- Academicians and researchers
- Students of all Departments

# WHY ATTEND

- Learn from experts and industry persons
- Gain knowledge with real-world applications
- Networking with professionals in the field

## REGISTRATION LINK

https://forms.gle/cFHKjGLonNiXDJx2A

DESIGN DECODE DELIVER
Mapping Microstructure to Performance from Atoms to Applications

# COURSE OUTLINE

## MODULES

### 01 DESIGN: ADVANCED ALLOYS

- Lightweight & Next-Gen Alloy Design
- High entropy & Complex Concentrated Alloys
- Advanced High-Strength & Microalloyed Steels
- Metallic Glasses & Amorphous Alloy

### 02 PROCESS: MICROSTRUCTURE BY DESIGN

- Thermomechanical Processing & Heat Treatment
- Additive Manufacturing of Alloys
- Severe Plastic Deformation & Grain Refinement
- Process-Structure-Property Relationships

### 03 DECODE: MICROSTRUCTURAL ASSESSMENT

- Phase Identification & Texture Analysis
- Grain Structure, Precipitates & Defects
- Characterization of Damaged Materials
- Live Demonstrations & Data Interpretation

### 04 DELIVER: PERFORMANCE & FAILURE

- Strength-Ductility & Deformation Mechanisms
- Fatigue, Creep, Fracture & Hydrogen. Embrittlement
- Damage & Defect Quantification
- Industrial Failure Case Studies
- Corrosion and surface degradation

### 05 PREDICT: MODELLING & SUSTAINABLE FUTURE

- CALPHAD, Phase-Field & Crystal Plasticity Modelling
- Machine Learning & Materials Informatics
- Simulation for Structural Integrity Assessment

## REGISTRATION FEE

- **External academia (B.Tech.) :** ₹295 (GST-included)
- **External academia (M.Tech./Ph.D.) :** ₹354 (GST included)
- **External academia (Faculties from institutes):** ₹590 (GST included)
- **Industry/Laboratory :** ₹1,180 (GST included)
- **Internal academia (B.Tech./M.Tech./Ph.D./Faculty) :** Fees waived

**NOTE : Participants who complete the course receive a Certificate of Participation**

## PAYMENT DETAILS

Account Number : 38246478714,
IFSC Code : SBIN0001882,
Account Name : RESEARCH AND CONSULTANCY NIT JAMSHEDPUR
Bank Name : State Bank of India (SBI)
Branch : SBI NIT Jamshedpur

## WEBSITE LINK

https://metamap-nitjsr.vercel.app/

CHIEF PATRON



PROF. (DR.) GOUTAM SUTRADHAR
DIRECTOR

PATRON



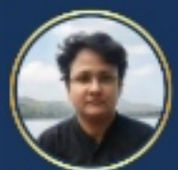
PROF. ASHOK KUMAR
DEAN SW

CHAIRMAN



DR. C.S. CHOWDHARY
HOD, MME DEPT.

CONVENORS



DR. ANUSHREE DUTTA
ASSISTANT PROFESSOR



DR. AMIT PATEL
ASSISTANT PROFESSOR

COORDINATORS



DR. PRAKASH SARKAR
ASSISTANT PROFESSOR



DR. JICHIL MAJHI
ASSISTANT PROFESSOR

## ORGANISING MEMBERS

FACULTY MEMBERS OF DMME, NIT JAMSHEDPUR: DR. RINA SAHU, DR. S K VAJPAI, DR. ARAVIND GALI, DR. POULAMI MAJI, DR. RENU KUMARI, DR. RAM KRISHNA, DR. PARTHA DULEY, DR. MONALISA MANDAL, DR. BINAY KUMAR, DR. ASHWINEE KUMAR, DR. SOUMITRA TARAFDAR, DR. MANISH ROY, DR. T.K. ROY